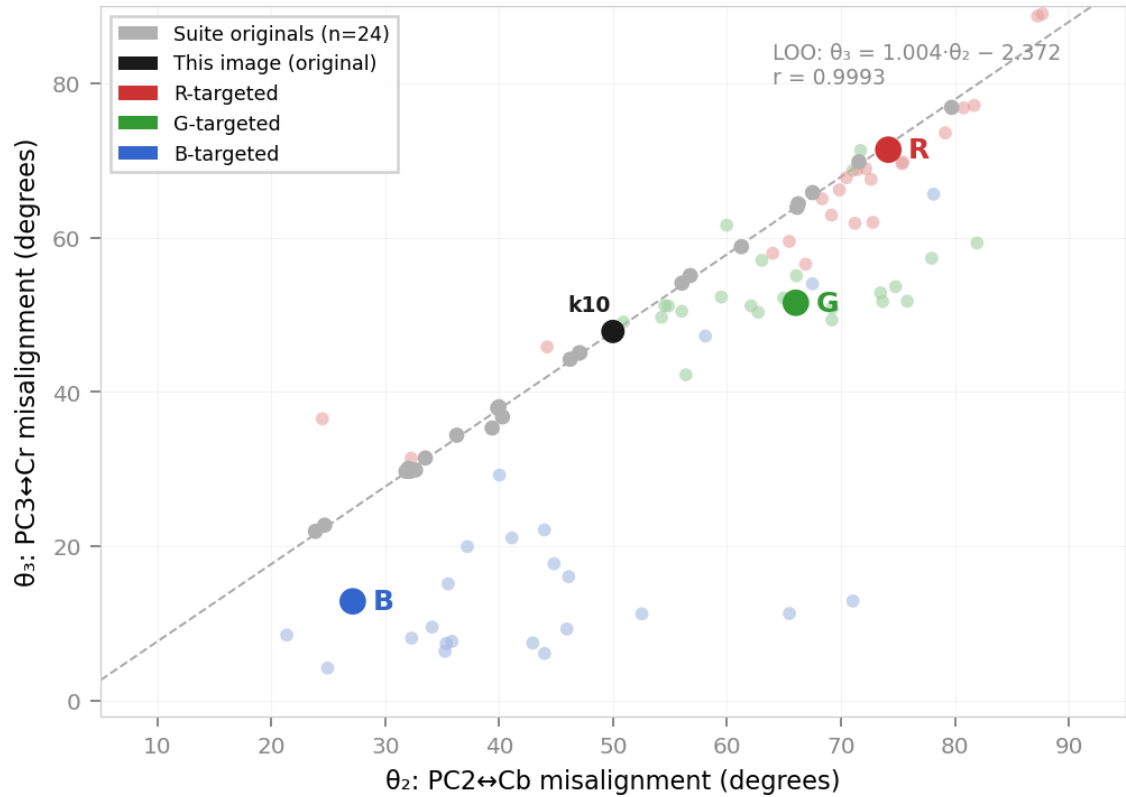


Compression Profile: KODIM10 512x768 | 24-bit RGB | PNG (lossless)

off-shore sailboat race | Strongly One-Dimensional (PC1 = 96.12%)

1. Geometric Displacement Under Directional Perturbation



2. Pairwise Correlation and Angular Misalignment

Version	R-G	R-B	G-B	Avg r	theta2	theta3	L00 dev
Original	0.9555	0.8902	0.9695	0.938	49.98°	47.86°	+0.05°
R-targeted	0.2074	0.2175	0.9693	0.465	74.16°	71.43°	-0.65°
G-targeted	0.1145	0.9247	0.1110	0.383	66.04°	51.58°	-12.35°
B-targeted	0.9125	-0.0032	0.1436	0.353	27.14°	12.86°	-12.02°

R-targeted isolates red (R-G, R-B < 0.22) while preserving G-B coupling at 0.969.
G-targeted isolates green (R-G, G-B < 0.11) while preserving R-B at 0.925.
B-targeted preserves R-G at 0.913 while driving R-B and G-B near zero.
All three perturbation directions depart from the L00 constraint manifold ($r = 0.9993$).

Compression Profile: KODIM10512x768 | 24-bit
RGB | PNG (lossless)

off-shore sailboat race | Strongly One-Dimensional (PC1 = 96.12%)

3. File Format and Size Comparison

All versions depict the same scene. BPP = (file size in bits) / (pixel count).

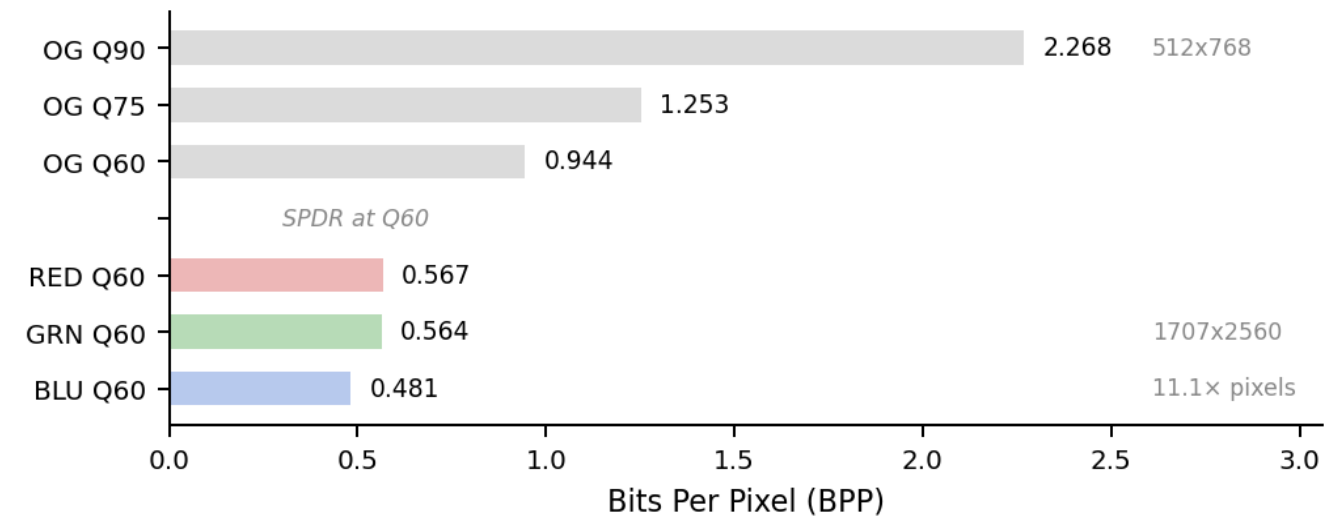
ORIGINAL (kodim10)

Version	Format	Resolution	Size	BPP	Pixels
PNG (lossless)	PNG	512x768	579.6 KB	12.074	393,216
JPEG Q90	JPEG	512x768	108.9 KB	2.268	393,216
JPEG Q75	JPEG	512x768	60.1 KB	1.253	393,216
JPEG Q60	JPEG	512x768	45.3 KB	0.944	393,216

REDISTRIBUTED (kodim10 SPDR at Q60)

Version	Format	Resolution	Size	BPP	Pixels
RED Q60	JPEG	1707x2560	302.7 KB	0.567	4,369,920
GRN Q60	JPEG	1707x2560	300.8 KB	0.564	4,369,920
BLU Q60	JPEG	1707x2560	256.6 KB	0.481	4,369,920

4. Quality Ladder: SPDR Q60 vs Original Q90



All three perturbation directions at Q60 produced lower BPP than the original at Q90, while carrying 11.1x the original pixel count.

Compression Profile: KODIM10 512x768 | 24-bit RGB | PNG (lossless)

off-shore sailboat race | Strongly One-Dimensional (PC1 = 96.12%)

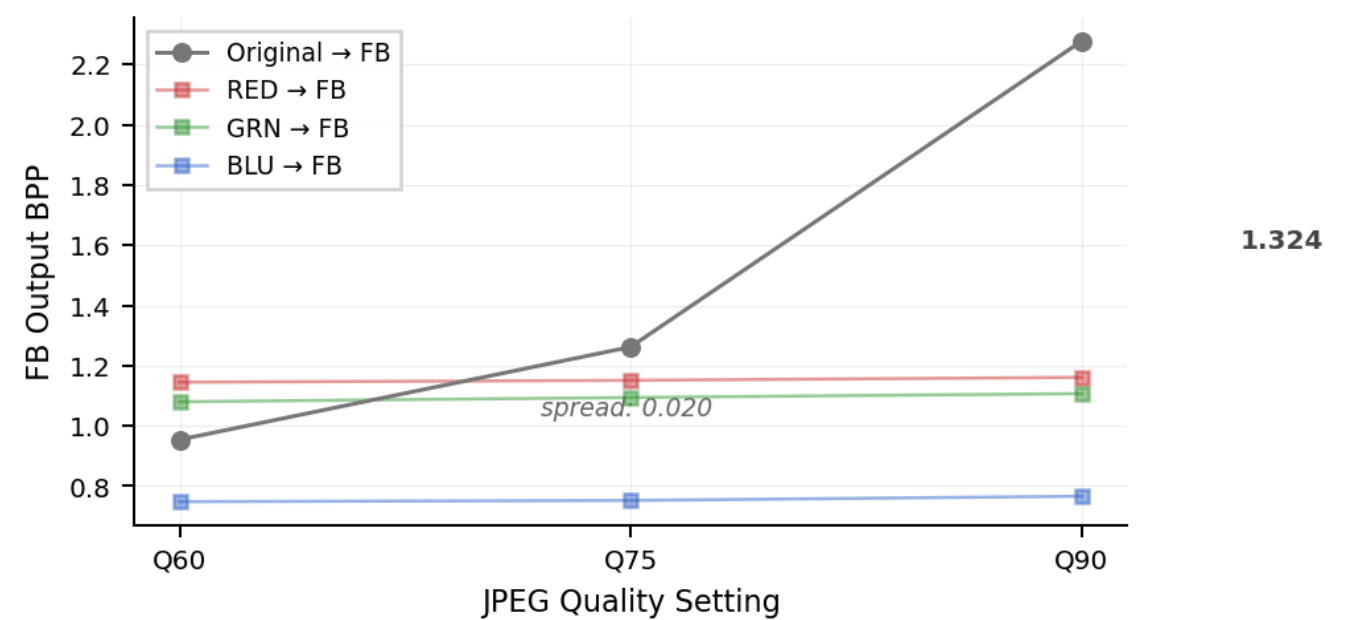
5. Third-Party Compression Pipeline (Facebook/Meta)

Each version uploaded to Facebook. Platform-compressed output measured at steady state (FB2).

Pipeline	Input Res	Output Res	FB BPP	BPP reduction
OG PNG → FB	512x768	512x768	2.122	—
R-targeted → FB	1707x2560	1366x2048	1.161	45.3%
G-targeted → FB	1707x2560	1366x2048	1.107	47.8%
B-targeted → FB	1707x2560	1366x2048	0.766	63.9%
ABC (no perturbation) → FB	1707x2560	1366x2048	1.179	44.4%

ABC control: same pipeline, same resolution, no covariance perturbation. Avg |r| unchanged (0.9379 vs 0.938 original). Comparable BPP reduction was not observed in the ABC control.

6. Quality Parameter Sensitivity Through Facebook



Original: Q60 → Q90 produces a 1.324 BPP spread through Facebook.
Redistributed: Q60 → Q90 produces a 0.020 BPP spread through Facebook.
Redistributed versions produced substantially smaller Q60-90 BPP variation through the measured pipeline.

7. Pipeline Summary – kodim10

Best BPP (Q60, direct)	0.481	1707x2560, 4,369,920 px
Best BPP through FB	0.766	1366x2048 output
vs OG Q90 → FB	66.4%	SPDR Q60→FB vs OG Q90→FB
Pixel ratio	11.1×	4,369,920 vs 393,216